

Progress Report

A Unified Benchmark of Grokking Predictors in Neural Networks

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Repository	grokking-benchmark (local git repository)

1. Current Status

The thesis empirically compares nine published grokking predictors on a canonical modular-arithmetic grokking task. As of this report, four of the nine predictors have been implemented, benchmarked over five random seeds, and formally closed with a negative result. The fifth predictor, HTSR-Alpha, has not yet been started.

The project timeline (Gantt chart, approved at proposal stage) set a milestone of “all nine predictors verified” by the end of the September Week 4 checkpoint. With 4 of 9 predictors closed and the fifth not yet started, the project is presently behind that checkpoint. The two milestones already passed — supervision agreement uploaded, and grokking reproduction achieved — were both met on schedule (reproduction gate delivered July 28, 2026, against a target of end of July).

Milestone	Target	Status
Supervision agreement uploaded	Jun, Week 4	Done
Grokking reproduction (gate)	Jul, Week 4	Done (Jul 28, 2026)
Baseline results	Aug, Week 4	Done
All 9 predictors verified	Sep, Week 4	In progress — 4 of 9 closed
Leaderboard and plots complete	Oct, Week 4	Not started
Ensemble and anti-grokking study	Nov, Week 3	Not started
Thesis submitted	Nov, Week 4	Not started

Table 1. Gantt milestones and current status.

2. Phase 1 Gate: Grokking Reproduction

Before comparing predictors, the canonical grokking phenomenon itself was reproduced, as a gate condition on the thesis proceeding. A single-head transformer was trained on the task $(a + b) \bmod 97$ for 10,000 epochs, and per-epoch loss and accuracy were logged.

Training accuracy reached 1.0000 by approximately epoch 935, that is, the network memorised the training set quickly. Test accuracy remained low for a long plateau, then crossed 0.50 around epoch 3406, and crossed 0.90 around epoch 4271. By the final epoch (10,000), test accuracy reached 0.9968, with training loss reduced to 0.0001. This is the standard grokking signature: rapid memorisation, followed by a long plateau, followed by a sharp jump to near-perfect generalisation. This gate was met on July 28, 2026.

3. Main Benchmark: Methodology (Nanda-Unified Protocol, v4)

The main benchmark reproduces the architecture and training protocol of Nanda et al. (2023), “Progress Measures for Grokking via Mechanistic Interpretability” (ICLR 2023), so that predictor behaviour can be compared against a well-characterised reference setting.

- Task: $(a + b) \bmod 113$, with vocabulary size 114.
- Architecture: 4-head transformer, model dimension 128, no LayerNorm, no linear biases.
- Optimiser: AdamW, learning rate 0.001, betas (0.9, 0.98), weight decay 1.0.
- Initialisation: small-init scheme (approx. standard deviation 0.0707), matching Nanda's TransformerLens setup.
- Training: full-batch, batch size 3830 (30% of all 113^2 input pairs), 40,000 epochs.
- Logging: 24 log-uniformly spaced checkpoints per run.
- Replication: 5 independent random seeds per predictor.

Across all 5 seeds, the network grokked in every run (5 of 5), with no pathological non-convergent (“limit-cycle”) runs. The grok epoch, defined as the epoch at which test accuracy first exceeds the chosen threshold, varied considerably by seed: 14474, 6988, 10418, 10193, and 24021, giving a mean of 13218.8 epochs with a standard deviation of 5900.6 epochs. This large seed-to-seed spread is itself a relevant finding: it means that any candidate predictor must track a moving target, not a fixed epoch.

4. Predictor Results (4 of 9 Complete)

Each predictor is evaluated on a single, pre-registered criterion: does the predictor’s signal reach its extremum (peak or crossover) before the grok epoch, consistently across all 5 seeds? A predictor is “closed negative” when the answer is no. This is a genuine falsification, not an inconclusive result: each predictor was implemented faithfully to its source paper before being tested.

#	Predictor	Source	Observed behaviour	Verdict
1	L2 Norm	Nanda et al. 2023 (Fig. 7)	Moving-average crossover fires very early (epochs 103–121 across seeds), thousands of epochs before grok in 5/5 runs.	Closed — negative
2	Dropout (variance)	Salah & Yevick, arXiv:2507.11645	Peak epoch lands at or after the grok epoch in 5/5 runs (peak/grok ratio 1.05–3.92).	Closed — negative
3	Spectral (task–model alignment)	Canatar, Bordelon & Pehlevan 2021	No pre-grok spectral concentration observed; representation entropy stayed flat ($8.24 \rightarrow 7.94$); alignment peak follows grok in 5/5 runs.	Closed — negative
4	AGE (Neural Collapse, NC1)	Papayan et al. 2020 (PNAS)	NC1 does collapse sharply (about 660-fold, from roughly 45 to 0.05) in all 5 seeds, but this collapse lags the grok epoch (ratio 1.67–3.92) rather than leading it.	Closed — negative
5	HTSR Alpha	Martin & Mahoney (weightwatcher)	Not yet started.	Pending
6	Correlation Traps	—	Not yet started.	Pending
7	Weight-PCA	—	Not yet started.	Pending
8	Higher-order Mutual Information	—	Not yet started.	Pending
9	Commutator Defect	—	Not yet started.	Pending

Table 2. Status of all nine predictors under investigation. Rows 1–4 report completed, falsified results from the Nanda-Unified v4 benchmark (git commit 680a108, tag benchmark-v4-4predictors, September 7, 2026). Rows 5–9 are not yet started.

A secondary, earlier finding is also on record: adding LayerNorm to the architecture, under the same weight-decay-1.0 regime, destabilised training. This variant was tested and not adopted; it is retained in the repository as a documented negative result rather than discarded.

5. Interpretation

The pattern across the four closed predictors is informative in itself. L2 Norm fails in the opposite direction from the other three: it fires far too early, while Dropout-variance, Spectral alignment, and AGE all fire at or after the grok event. This suggests that no single one of these four signals, taken in isolation, can serve as an early warning of generalisation. Whether some combination of the remaining five predictors, or an ensemble of all nine, performs better is exactly the question the later phases of the thesis (leaderboard construction and ensemble study) are designed to answer.

6. Next Steps

- Implement Predictor 5 (HTSR-Alpha, Martin & Mahoney), using the weightwatcher library already listed as a project dependency.
- Following the established cadence, implement and formally close Predictors 6–9 (Correlation Traps, Weight-PCA, Higher-order Mutual Information, Commutator Defect).
- Given the delay against the September Week 4 milestone, revise the remaining schedule with the supervisor and confirm whether the October Week 4 (leaderboard and plots) and November (ensemble, anti-grokking, submission) milestones remain feasible.

Source data: results/nanda_unified/aggregate.json and results/nanda_unified/BENCHMARK-V4.md (git commit 680a108);
Phase-1 gate log: docs/table_raw.txt.